Near-Field Communication with Movable Antennas: An Equilibrium Perspective

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GitHub Toy Demo

Brief Summary

Optimal antenna placement in near-field LoS wirelss communications for SE maximization [1], [2].

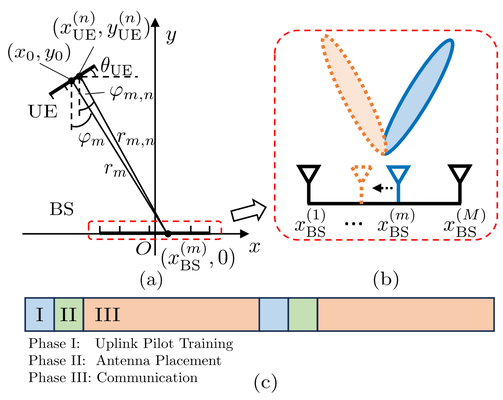
- **Equilibrium** positions with two fixed charges and one external charge.
  - **Greedy** [1]: Approximation via **Leja sequence**
  - **Closed-form** [2]: Roots of polynomial solutions to Stieltjes ODE.

1. Motivations

- **Near field:** Spherical wavefronts over large apertures expose higher spatial DoF that ULA cannot fully utilize.
- **Position Reconfigurable Antennas (MA / FAS / PASS)** sample channel non-uniformly. Utilize DoF by changing the intrinsic channel property.
- Existing placement schemes lacks **Physical Insights** and mostly rely on numerical optimization with **high complexity**.

Goal: A low-complexity, physically interpretable placement rule that unlocks the near-field spatial DoF.

2. System Model



- Aperture $A_{\mathrm{T}} = (M-1)d$, antenna position (ascending order):

$$-\frac{A_{\mathrm{T}}}{2} = x_{\mathrm{BS}}^{(1)} < x_{\mathrm{BS}}^{(2)} < \cdots < x_{\mathrm{BS}}^{(M)} = \frac{A_{\mathrm{T}}}{2}, \tag{1}$$

- Near-field LoS channel

$$\boldsymbol{H}_{\boldsymbol{x}}[n,m] = \frac{e^{\jmath\kappa r_{m,n}}}{r_{m,n}} \tag{2}$$

- SE objective

$$R_{\boldsymbol{x}} = \log_2 \det\left(\boldsymbol{I}_N + \frac{\rho}{M}\boldsymbol{H}_{\boldsymbol{x}}\boldsymbol{H}_{\boldsymbol{x}}^H\right). \tag{3}$$

3. Problem Formulation

- For precision, apply **Trigonometric substitution**

$$\begin{aligned} s_m &= \sin(\varphi_m - \theta_{\mathrm{UE}}) = \sin\left(\arctan\left(\frac{x_{\mathrm{BS}}^{(m)} - x_0}{y_0}\right) - \theta_{\mathrm{UE}}\right) \\ &= \sin(\varphi_m - \theta_{\mathrm{UE}}) \end{aligned} \tag{4}$$

- SE maximization problem transformed into a **Weighted Fekete Problem** (WFP) [3]

$$\mathcal{P}_2: \begin{array}{ll} \max\limits_{\boldsymbol{s}} & 2\sum\limits_{1\le i<j\le M}\log_2(s_j - s_i) + \sum\limits_{m=1}^{M}\log_2(1-\tilde{s}_m^2) \\ \text{s.t.} & s_{\min} = s_1 < s_2 < \cdots < s_M = s_{\max}, \\ & \tilde{s} = \sin(\arcsin s + \theta_{\mathrm{UE}}). \end{array} \tag{5}$$

Convex and elegant. With lots of target variables, generic interior-point solvers could be slow and give *limited design insight*.

4. Leja Greedy Design [1]

- Place antenna one by one, each at the maximum of the incremental Fekete objective

$$\begin{aligned} J_{\mathrm{D}}(s) &= 2\sum_{i=1}^{m-1}\log_2|s_i^\star - s| \\ J_{\mathrm{W}}(s) &= \log_2(1-\sin^2(\arcsin s + \theta_{\mathrm{UE}})). \end{aligned} \tag{6}$$

- Derivative-free, low-complexity, near-optimal.

Greedy Antenna Position Design

Input: M, spacing thresholds ϵ, aperture A, UE position (x_0, y_0) and rotation θ_{UE}.
Output: Positions $\{x_{\mathrm{BS}}^{(m)}\}_{m=1}^M$ of antennas.
1: Fix endpoints $s_1^\star, s_M^\star$ at the aperture boundaries
2: **for** $m = 3, \cdots, M$ **do**
3: Update the feasible region

$$\mathcal{S}_m = \{s \in \mathbb{R} \mid s_1 < s < s_M\} \setminus \bigcup_{i=1}^{m-1} B_\epsilon,$$

4: Determine the m-th position over $\mathcal{S}_m$

$$s_m^\star = \operatorname*{argmax}_{s\in\mathcal{S}_m}\left(J_{\mathrm{D}}(s) + J_{\mathrm{W}}(s)\right)$$

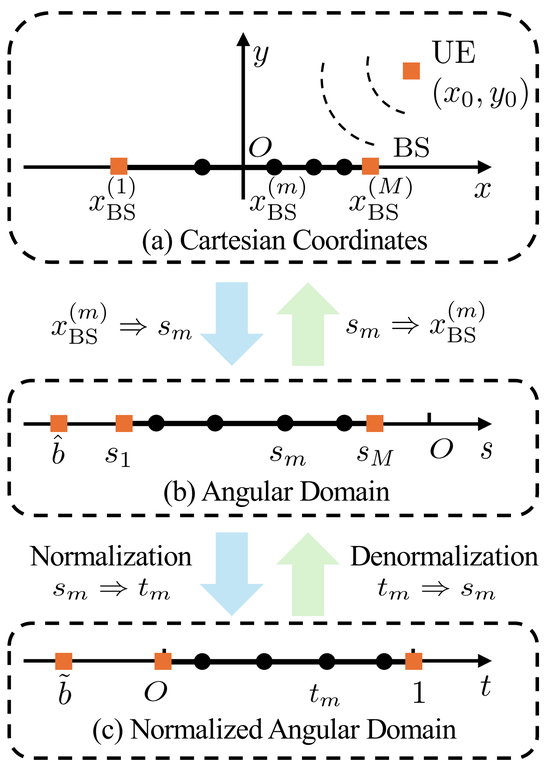
5: **end for**
6: Angular positions $\{s_m^\star\}_{m=1}^M$ to coordinates $\{x_{\mathrm{BS}}^{(m)}\}_{m=1}^M$.

- Execution time [1]

Methods	Computational Complexity ($\mathcal{O}$)	Ave. Exec. Time (ms) $M=16$	Ave. Exec. Time (ms) $M=128$
AS	$MG^2(N^3+M^2N^2)$	47.4	3.38×10^3
Prop. CVX	$\sim M^{4.5}$	6.42×10^3	3.47×10^5
Prop. Greedy	GM^2	**2.19**	**94.1**

* Using interior-point method.

5. Electrostatic Equilibrium [2]



- First-order optimality of $\mathcal{P}_2$:

$$\underbrace{\sum_{\substack{i=1\\ i\neq m}}^{M}\frac{2}{s_m - s_i}}_{\textbf{internal force}} = \underbrace{\frac{2\tan\varphi_m}{\sqrt{1-s_m^2}}}_{\textbf{external force}} \simeq \frac{b}{a - b(s_m - s_0)}, \tag{7}$$

where $s_0 = \sin(\varphi_0 - \theta_{\mathrm{UE}})$, $a = \cos^2\varphi_0$, and $b = \sin(2\varphi_0)/\cos(\varphi_0 - \theta_{\mathrm{UE}})$.

- Apply normalization $t = (s - s_{\min})/(s_{\max} - s_{\min})$

$$\sum_{\substack{i=2\\ i\neq m}}^{M-1}\frac{2}{t_m - t_i} = -\frac{2}{t_m} - \frac{2}{t_m - 1} - \frac{1}{t_m - \tilde{b}}, \tag{8}$$

This is exactly the equilibrium of $M-2$ free charges within $(0,1)$ repelling each other, with **three fixed charges** at 0, 1, and $\tilde{b}$.

- The optimal placement is the *electrostatic equilibrium* of the charge system.

Physical analogy: M charges on a line, mutually repelling yet pulled by an external charge (the UE) into *equilibrium*.

6. Two-Step EVD-based Solution

- Analytical solution to equilibrium: Polynomial $p(t)$ roots $\{t_m\}_{m=1}^{M-1}$ of Stieltjes ODE [4].

$$\frac{p''(t_m)}{p'(t_m)} = \sum_{\substack{i=2\\ i\neq m}}^{M-1}\frac{2}{t_m - t_i} \quad\Rightarrow\quad A(t)p'' + B(t)p' + V(t)p = 0, \tag{9}$$

with $A = t(t-1)(t-\tilde{b})$, $B = 2(t-1)(t-\tilde{b}) + 2t(t-\tilde{b}) + t(t-1)$, and $V = v_1 t + v_0$.

- With only one LoS path, this is the canonical **Heun equation**:

$$p''(t) + \left(\frac{\gamma}{t} + \frac{\delta}{t-1} + \frac{\epsilon}{t-a_{\mathrm{H}}}\right)p'(t) + \frac{\alpha\beta t - \lambda}{t(t-1)(t-a_{\mathrm{H}})}p(t) = 0,$$

with $\gamma = \delta = 2$, $\epsilon = 1$, $a_{\mathrm{H}} = \tilde{b}$, $\alpha = -(M-2)$, $\beta = M+2$, and $v_1 = \alpha\beta = -(M-2)(M+2)$.

- **EVD Step I**: Build matrix $\boldsymbol{R}$ and obtain its eigenvector $\boldsymbol{q}$

$$\boldsymbol{R} = \begin{bmatrix} B_0 & C_0 & 0 & \cdots & 0 \\ A_1 & B_1 & C_1 & \ddots & \vdots \\ 0 & A_2 & B_2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & C_{M-2} \\ 0 & \cdots & 0 & A_{M-2} & B_{M-2} \end{bmatrix} \Rightarrow \boldsymbol{q} \tag{10}$$

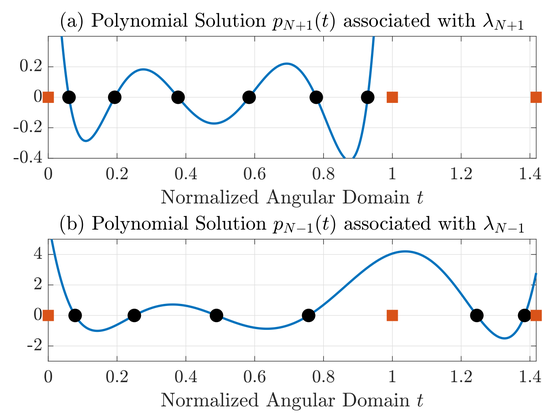
where

$$\begin{cases} A_n = (n-1+\alpha)(n-1+\beta) \\ B_n = -n\left((n-1+\gamma)(1+a_{\mathrm{H}}) + a_{\mathrm{H}}\delta + \epsilon\right). \\ C_n = (n+1)(n+\gamma)a_{\mathrm{H}} \end{cases} \tag{11}$$

- **EVD Step II**: Build matrix $\boldsymbol{C}$ with $\boldsymbol{q}$

$$\boldsymbol{C} = \begin{bmatrix} \boldsymbol{0}_{1\times(M-3)} & -\frac{\boldsymbol{q}_{0:M-3}}{\boldsymbol{q}_{M-2}} \\ \boldsymbol{I}_{M-3} & \end{bmatrix} \in \mathbb{R}^{(M-2)\times(M-2)} \Rightarrow \boxed{\text{Zeros of } p(t)} \tag{12}$$

- Selecting the desired eigenvector $\boldsymbol{q}$ is important. Different $\boldsymbol{q}$ leads to different solutions



- We pick $\boldsymbol{q}$ according to Sturm-Liouville theory [5]
  - $\tilde{b} < 0$: pick the **largest** eigenvalue λ_{M-1};
  - $\tilde{b} > 1$: pick the **smallest** eigenvalue λ_1;
  - $0 < \tilde{b} < 1$ (rare): evaluate all $M-1$ eigenvectors and keep the one maximizing $J(\boldsymbol{s})$.

7. Asymptotic Closed Form

- As $M \to \infty$ in the near-field region, the internal repulsion dominates

$$\begin{aligned} &\sum_{1\le i<j\le M}\log_2(s_j - s_i) \gg \sum_{m=1}^{M}\log_2(1-\tilde{s}_m^2) \\ \Rightarrow\quad &\sum_{\substack{i=2\\ i\neq m}}^{M-1}\frac{1}{s_m - s_i} + \frac{1}{s_m - s_{\min}} + \frac{1}{s_m - s_{\max}} = 0, \end{aligned} \tag{13}$$

- The equilibrium condition reduces to the **Jacobi equation**

$$\sum_{\substack{i=2\\ i\neq m}}^{M-1}\frac{1}{t_m - t_i} = -\frac{1}{t_m + 1} - \frac{1}{t_m - 1} \quad\Rightarrow\quad (1-t^2)p''(t) - 4tp'(t) + v_0 p(t) = 0, \tag{14}$$

and the exact solution is the **Jacobi Polynomial** $p(t) = P_{M-2}^{(1,1)}(t)$.

- As $M \to \infty$, it admits a closed form (a.k.a., Chebyshev nodes)

$$t_n = \cos\left(\frac{(2n+1)\pi}{2(M-2)}\right), \quad n = 0, \ldots, M-3, \qquad \rho(t) = \frac{1}{\pi\sqrt{1-t^2}}. \tag{15}$$

- Transform back to coordinates
  - Denormalize

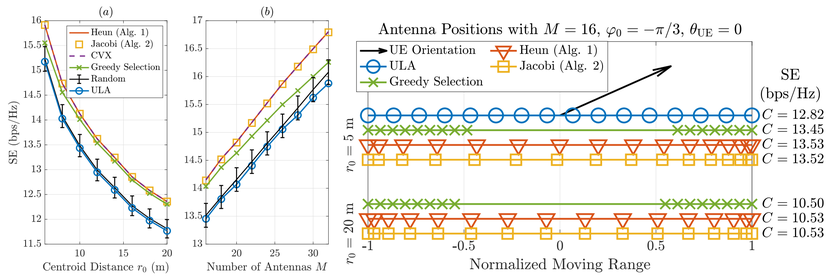
$$s_m = t_m(s_{\max} - s_{\min}) + s_{\min} \tag{16}$$

  - Inverse transform

$$x_{\mathrm{BS}}^{(m)} = x_0 + y_0\tan(\theta_{\mathrm{UE}} + \arcsin s_m) \tag{17}$$

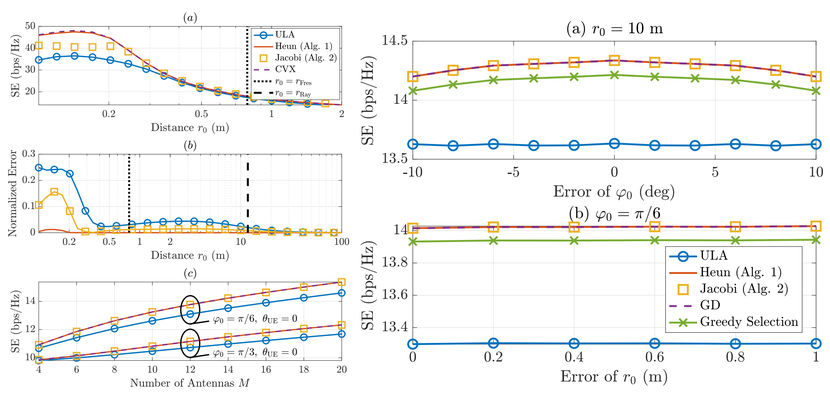
8. Numerical Results

- Setup: $f_{\mathrm{c}} = 10$ GHz, $M = 16$, $N = 8$, spacing $d = 2\lambda$, SNR $= 20$ dB.
- (Left) SE performance with varying distance/number of antennas; (Right) Robustness



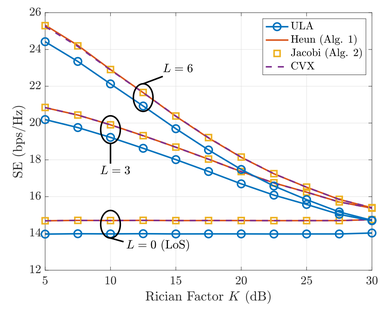
EVD and closed form coincide with **CVX**; ∼5–6% gain over ULA.
Dense at the edges, sparse at the center; Elements cluster toward the near UE side.

- Asymptotic accuracy



Normalized error ≤ 5% within $r_{\mathrm{Fres}} \le r_0 \le r_{\mathrm{Ray}}$; closed form matches EVD from $M = 4$.
10° angle error costs 0.96% SE; Distance mismatch is almost harmless.

- Multipath performance



Heun and Jacobi remain close to **CVX** across Rician factors and path numbers, while consistently outperforming ULA.

- Complexity table

Method	Complexity	Avg. time (s) $M=16$	Avg. time (s) $M=64$
EVD (2-step)	$2M^3 + 5M$	7.4×10^{-5}	9.1×10^{-2}
Closed form	—	$\mathbf{2.6\times10^{-7}}$	$\mathbf{4.1\times10^{-7}}$
CVX	$\sim M^3$	5.2×10^{-2}	2.74
Greedy	$\mathcal{O}(MN^3)$	1.1×10^{-2}	2.2×10^{-1}

EVD: ∼30× faster than CVX at $M = 64$.

Take-Home Messages

- Near-field MA placement is an **electrostatic equilibrium** problem.
- The solution is obtained by performing two-step EVD
- Jacobi approximation shows insignificant performance loss with a closed-form solution.
- The same picture extends to multiuser/ISAC (more external charges) and planar arrays (future works).

References

[1] S. Liu and X. Yu, "Near-field line-of-sight communication with massive movable antennas," in *IEEE International Conference on Communications (ICC'26)*, Glasgow, Scotland, UK, 2026, pp. 1–6.
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[4] M. Marden, "On Stieltjes polynomials," *Trans. Am. Math. Soc.*, vol. 33, no. 4, pp. 934–944, Oct. 1931, ISSN: 00029947, 10886850.
[5] M. A. Al-Gwaiz, *Sturm-Liouville Theory and its Applications* (Springer Undergraduate Mathematics Series). Springer London, 2008, ISBN: 978-1-84628-971-2.